

Progress Test 3.C — Elec-S1

3.C · Operating point of a circuit

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Maximum power

(1 mark)

A source has $E_{Th} = 12\text{ V}$ and $R_{Th} = 6.0\ \Omega$. Give the maximum power it can deliver to a resistive load.

Matching $R_c = R_{Th}$:	$P_c^{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{144}{4 \times 6.0}\text{ W}$	$P_c^{\max} = 6.0\text{ W}$
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Marking

①
0 1

Marking scheme

① 1 pt $P_c^{\max} = 6.0\text{ W}$.

Exercise 2. The operating point

(5 marks)

A linear network, described by its Thévenin model (E_{Th}, R_{Th}), feeds a load resistor R_c . Define the operating point. Write down the characteristic of the source and that of the load, and deduce the analytical expressions of u^* and i^* .

The operating point is the actual state of the circuit: the unique pair (u^*, i^*) that satisfies <i>simultaneously</i> the characteristic of the source and that of the load. Graphically, it is the intersection of the two characteristics. Source (generator convention): $u = E_{Th} - R_{Th} i$ Load (receiver convention): $u = R_c i$ $R_c i^* = E_{Th} - R_{Th} i^* \Rightarrow i^* = \frac{E_{Th}}{R_{Th} + R_c}$ $u^* = R_c i^* = \frac{R_c}{R_{Th} + R_c} E_{Th}$	
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Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① **2 pts** Pair (u^*, i^*) satisfying both characteristics (1 pt); intersection of the two curves (1 pt).
- ② **2 pts** $u = E_{Th} - R_{Th}i$ (1 pt) and $u = R_c i$ (1 pt).
- ③ **1 pt** Both i^* and u^* .

Exercise 3. Power transfer

(3 marks)

When the load resistance tends to infinity (open circuit):

- | | |
|---|--|
| ☐ $u^* \rightarrow 0$ and $i^* \rightarrow E_{Th}/R_{Th}$ | ☒ $u^* \rightarrow E_{Th}$ and $i^* \rightarrow 0$ |
| ☐ $u^* \rightarrow E_{Th}$ and $i^* \rightarrow E_{Th}/R_{Th}$ | ☐ $u^* \rightarrow 0$ and $i^* \rightarrow 0$ |

Why: $i^* = E_{Th}/(R_{Th} + R_c) \rightarrow 0$ and $u^* \rightarrow E_{Th}$: the point reaches the voltage axis, and the load receives no power.

When the load is matched to the source ($R_c = R_{Th}$), the efficiency η of the transfer is:

- | | | | |
|--------------------------------|-------------------------------|--|-------------------------------|
| ☐ 100 % | ☐ 75 % | ☒ 50 % | ☐ 25 % |
|--------------------------------|-------------------------------|--|-------------------------------|

Why: $\eta = R_c/(R_{Th} + R_c) = 1/2$: as much power is lost in R_{Th} as reaches the load.

To transmit electric power over long distances, one chooses:

- ☐ matching, $R_c = R_{Th}$, to obtain the maximum power;
- ☒ a high efficiency, $R_c \gg R_{Th}$, to limit the losses;
- ☐ $R_c \ll R_{Th}$, to obtain the maximum current;
- ☐ a short circuit at the end of the line.

Why: energy is costly and losses cause heating, so efficiency wins. Matching is for tiny signals (antenna, sensor), where extracting the most power is what matters.

Exercise 4. Operating point, power and efficiency

(5 marks)

A source with $E_{Th} = 12\text{ V}$ and $R_{Th} = 4.0\ \Omega$ feeds a load $R_c = 8.0\ \Omega$. Determine the operating point (u^*, i^*), the power P_c received by the load and the efficiency η of the transfer. Check the power balance.

$E_{Th} = 12\text{ V}$	$R_{Th} = 4.0\ \Omega$	$R_c = 8.0\ \Omega$
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$i^* = \frac{E_{Th}}{R_{Th} + R_c} = \frac{12\text{ V}}{12\ \Omega}$	$i^* = 1.0\text{ A}$	$u^* = R_c i^*$	$u^* = 8.0\text{ V}$
$P_c = R_c i^{*2} = 8.0\ \Omega \times (1.0\text{ A})^2$	$P_c = 8.0\text{ W}$		
$\eta = \frac{P_c}{E_{Th} i^*} = \frac{R_c}{R_{Th} + R_c} = \frac{8.0}{12}$	$\eta = 2/3 \approx 67\%$		
Balance: produced by the ideal source $E_{Th} i^* = 12\text{ W}$ = lost in R_{Th} , $R_{Th} i^{*2} = 4.0\text{ W}$, + received by the load, 8.0 W . ✓			

Marking

①	0	1
	☐	☐
②	0	1
	☐	☐
③	0	1
	☐	☐
④	0	1
	☐	☐
⑤	0	1
	☐	☐

Marking scheme

- ① 1 pt $i^* = 1.0\text{ A}$.
- ② 1 pt $u^* = 8.0\text{ V}$.
- ③ 1 pt $P_c = 8.0\text{ W}$.
- ④ 1 pt $\eta = 2/3 \approx 67\%$.
- ⑤ 1 pt Balance $12\text{ W} = 4.0\text{ W} + 8.0\text{ W}$.

Examiner's tip

η is the ratio of the power received by the load to the power produced by the *ideal* source $E_{Th} i^*$, not to $P_c^{\max}$.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20